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2) LASSO Regression [L1]

alpha * abs(m)

$$Loss = MSE + \underbrace{\lambda \cdot \|w\|}_{\lambda\,[|w_1| + |w_2| + |w_3| + |w_4|]}$$

```
From sklearn.linear_model import Lasso
~~L = lasso~~ L = Lasso(alpha = 0.1)
L.fit(X_train, y_train)
y_pred = L.predict(X_test)
```

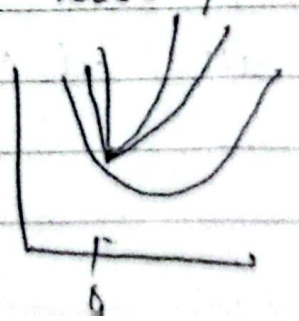
$\lambda \uparrow \rightarrow w \downarrow$ upto 0.

[But in Ridge we ~~reach~~ can't reach 0, only near to zero]

→ Advantage: It can reach zero.
In high dimension data. If the column is less imp, then corresponding λ will be zero.
It means indirectly, we are doing feature selection. [dimensionality decreases].
So, prefer Lasso

Effect in Loss f^n same as Ridge

But



final at zero.

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Q Why does Lasso Regression create sparsity?

[having zero value in model].
Most coef = 0
Few are non zero

$y = mx + b$

$$\left\{ \begin{aligned} b &= \bar{y} - m\bar{x} \\ m &= \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \end{aligned} \right\}$$ simple linear regression.

For Ridge Regression:

$$m = \frac{\sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^{n} (x_i - \bar{x})^2 + \lambda}$$

For Lasso Regression:

$b = \bar{y} - m\bar{x}$

$$L = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \lambda \cdot |m|$$

$$L = \sum_{i=1}^{n} (y_i - m x_i - \bar{y} + m\bar{x})^2 + \lambda |m|$$

we cant do diff $\frac{\partial L}{\partial m}$ coz $|m|$.

So we break m in diff interval.

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For $m > 0$

$$L = \sum_{i=1}^{n} (y_i - m x_i - \bar{y} + m\bar{x})^2 + 2\lambda \cdot m$$

2 is for mathematical convenience.

$$\frac{\partial L}{\partial m} = 0$$

then; $$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x}) - \lambda}{\sum (x_i - \bar{x})^2}$$

For $m = 0$

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$$

For $m < 0$

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x}) + \lambda}{\sum (x_i - \bar{x})^2}$$

For $m > 0$ & as always $\lambda > 0$

For simplicity: $$m = \frac{YX - \lambda}{X^2}$$

as $\lambda \uparrow$, $m \downarrow$

when $YX = \lambda$ } then $m = 0$.

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If $XY < h$, then m will be negative.
~~But~~ then we have to use formula of.

$$m = \frac{XY + h}{X^2} \quad [\because m < 0]$$

Let $m = 2$. at first $h = 1$.

m	2	1.0	0	5
h	1	10	50	100

make ~~more~~ model ~~ive~~ more degrade.
So it automatically stop in 0.

For $m < 0$:

$$m = \frac{XY + h}{X^2}$$

Here: $XY = \Sigma (y_i - \bar{y})(x_i - \bar{x})$
should be negative for $m < 0$

m	-2	-1	0	-5
h	0	50	100	150

After zero use $[m > 0]$ $\quad -\frac{100 - 150}{50}$

So stop in 0.

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(a) Why not sparsity in Ridge regression?

$$m = \frac{\Sigma (y_i - \bar{y})(x_i - \bar{x})}{\Sigma (x_i - \bar{x})^2 + \lambda}$$

Here; any value of λ can't make zero coz here λ is in denominator.

But In Lasso λ is in numerator.

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